

ATI

RINGLOCK AND TUBE & CLAMP

TECHNICAL MANUAL

NORTH AMERICA

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RINGLOCK AND TUBE & CLAMP
TECHNICAL MANUAL

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INTRODUCTION

ATI Products offered Complete Scaffolding Solutions to our strategic clients around the world, matching our expertise to the specific needs of our customers. With our extensive experience in the industrial market, we specialize in: Oil, Gas & Chemicals, Power & Energy, and Mining & Refining projects. ATI prides itself on building long-term and sustainable relationships with both our customers and team members. We only distribute the highest quality products by ensuring that our production processes adhere to our rigorous quality assurance and quality control programs.

We currently supply Asia-Pacific, Africa, the Middle East, and beyond. Our unparalleled expertise and professionalism enable us to deliver Complete Scaffolding Solutions consistently. Our experienced, customer-driven, team provides a seamless service from product development, to supply chain management, production, quality control, delivery and local engineering support for fast, effective solutions where required.

ATI Ringlock is a modular system scaffold that enables users to quickly and efficiently erect, use, and dismantle temporary work structures. The Ringlock system conforms to European, Chilean, Australian and ANSI/OSHA Regulation standards.

ENGINEERED WITH CONSISTENT QUALITY

All Ringlock products are engineered to meet the highest quality standards based on three simple factors:

MATERIAL Highest quality steel in our products. Steel is certified and tested to meet and exceed industry standards in any climate or locale.

FIT Products are engineered with versatility and usability in mind. Our modified product designs increase efficiency and productivity, saving your projects time and money.

FINISH Where applicable, our products are finished and protected by zinc galvanizing. This method of finishing maximizes the utilization and extends the product life.

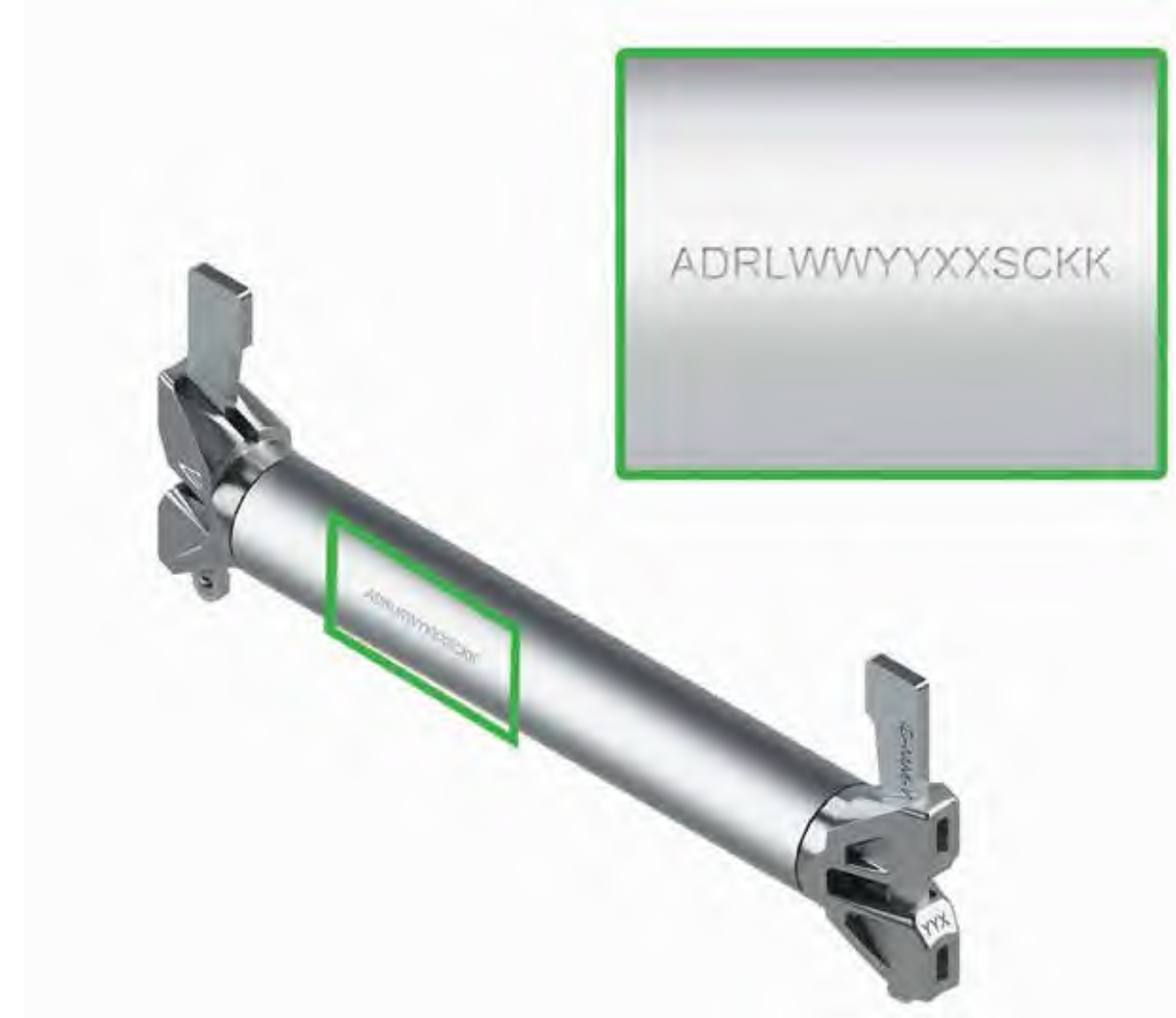
For additional information on Ringlock system components, please consult your local ATI representative.

All terms used in this document are synonymous with terms used in industry guidance.

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PRODUCT TRACEABILITY

All components are stamped with traceability marks.



Key

Batch Number: AD RL WW YY XX SC KK

AD = AT-PAC/DOKA
RL = Ringlock
WW= Week of Manufacture
YY = Year of Manufacture
XX = Order Number
SC = Reference Number
KK = Welding Member Identification

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Ringlock Connection Steps

RINGLOCK NODE CONNECTION

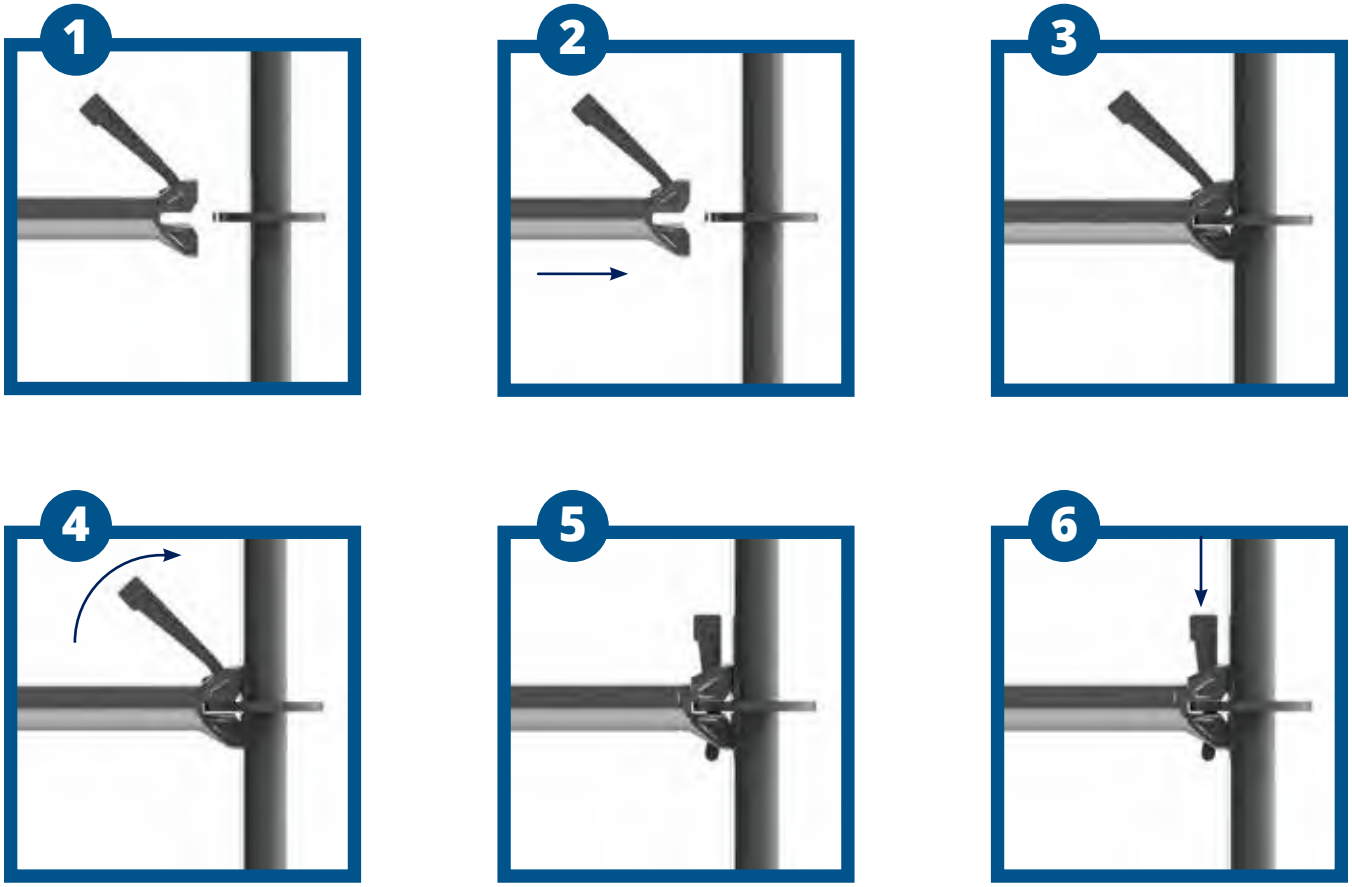
Ringlock node connection consists of Standards with rosettes every 19.7" (0.5m) and horizontal members with a cast end captive wedge (Ledger and Transom). When connected together the Ledger ends and rosettes create a very strong connection. This connection transmits the forces throughout the scaffold structure.

The Ringlock rosette has a diameter of 5” (123mm) and has been manufactured with 8-trapezoidal slots, 4-small and 4-large. The smaller holes represent the 90 degree angle at the Ledgers and Transoms are installed. The larger trapezoidal holes accommodate the Diagonal Bay Braces.

If the needed, the Ledgers can be installed in the larger holes, allowing up to 15-degrees of rotation. This provides added flexibility to clear obstacles when necessary.

It is recommended that a 500g (1.1lb) hammer is used to secure the Ringlock wedge, when the hammer rebounds, the wedge is secure.

TO MAKE A CONNECTION



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MK1 & MK2 RINGLOCK

The following values shown in this manual are 'Safe Working Capacities'. These are derived by dividing the Ultimate Strength by a Factor of Safety (F.O.S.) of 4. (Unless otherwise noted).

Since early 2015, RingLock has been manufacturing its Mk2 Ringlock System. The following data within this manual is based on Mk1 and Mk2 Ringlock Systems. Where appropriate, the lower of the Mk1 and Mk2 values have been used taking into account the possibility that both Ringlock systems may be used together.

The below illustrations show the difference between Mk1 & MK2 Ledger ends for identification purposes.

MK1 LEDGER HEAD:



MK2 LEDGER HEAD:



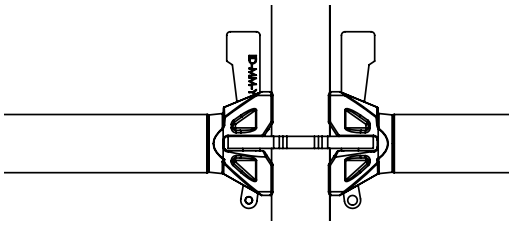
Note: Illustrations throughout this document reference MK2 components unless otherwise noted.

This document should be read in conjunction with Product Catalog and Ringlock Safety Manual & User Guide

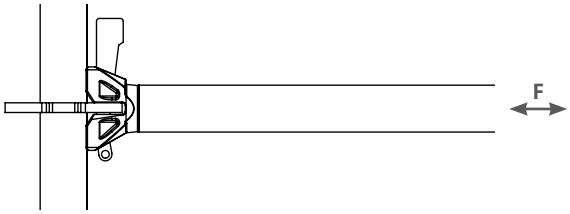
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RINGLOCK NODE CAPACITIES

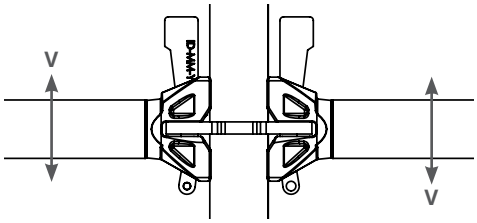
The following values and the values throughout this manual are 'Safe Working Capacities'. These are derived by the design load divided by the Safe Working Load Factor 4:1.



Connection Moment
(M+ = +228.5 lb-ft)
(M- = -298.8 lb-ft)



Horizontal Force O Type
(F= ± 3397.9 lb)



Vertical Force
(V = ± 1638.8 lb)

Test Report: 01 220 CHN/T-2305129 & 01 220 CHN/T-2305130 (F.O.S. 4:1)

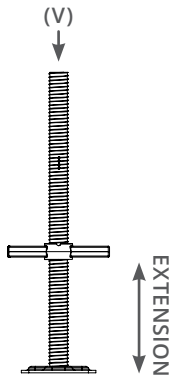
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Base

SCREW JACK/BASE JACK

VERTICAL LOAD			
EXTENSION		VERTICAL LOAD (V)	
in	mm	lb	kN
6	150	7885	35.09
12	305	7600	33.81
15	381	7075	31.48

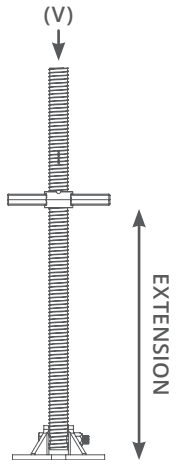
Test report: ATL083-10-06-72865-1. (F.O.S. 4:1)



SWIVEL JACK

VERTICAL LOAD			
EXTENSION		VERTICAL LOAD (V)	
in	mm	lb	kN
6	150	10339	46.00
12	305	7075	31.48
18	457	5980	26.60

Test report: ATL083-08-16-04407-1 & Test report: ATL083-12-23-45896-6. (F.O.S. 4:1)
***Note**: A sliding force occurs when on a slope. The capacity of the Swivel Screw Jack is dependent on the slope and the extent of the sliding force.*



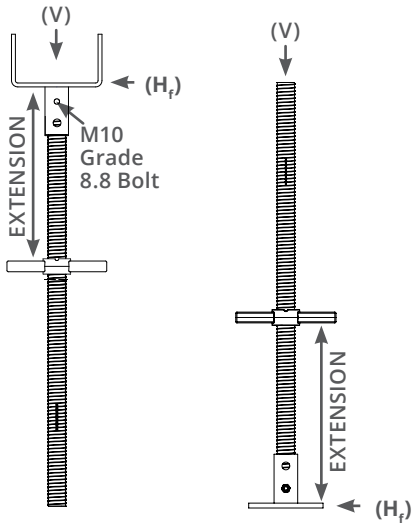
UNIVERSAL SCREW

UNIVERSAL SCREW JACK (0.86m)			
MAX EXTENSION		VERTICAL LOAD	
in	mm	lb	kN
24.4	620	9251	41.15

***Note**: Used with Base Plate:*
Base Plate must be erected on flat and even ground and must be erected plumb.
Suitable Sole Boards may be required as load distributing measure.
Ground must be able to safely support the leg loads from the scaffold.
H_f = Suitable lateral support at maximum spindle extension.
M10 Grade 8.8 bolt used to secure Universal Screw Jack to Unversal Base Plate.

Used With U-Head (150mm X 170mm):
M10 Grade 8.8 bolt used to secure Universal Screw Jack to U-Head.
H_f = Suitable lateral support at maximum spindle extension.

Test report: SHIN161204292CCM (F.O.S. 4:1).
Safe Working Vertical Load is based on maximum extension.

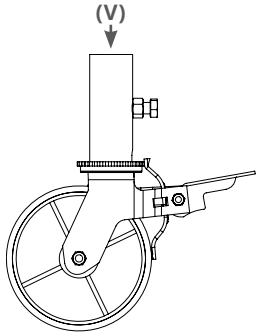


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CASTER 8”

STATIC LOAD			
DIAMETER		VERTICAL LOAD (V)	
in	mm	lb	kN
8	203	1323	5.89

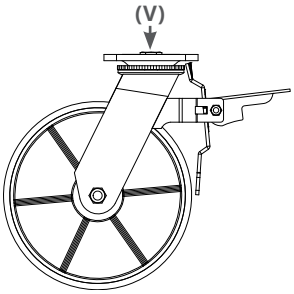
Test Report: 20122405M02. (F.O.S. 4:1)
***Note**: Caster must be locked*



CASTER 12”

STATIC LOAD				
DIAMETER		VERTICAL LOAD (V)		NOTE
in	mm	lb	kN	
12	305	2500	11.12	Attached to Caster Adapter
		2500	11.12	Attached to Caster Adjustable Base (6in Extension)

Test Report: ATL083-08-11-58779-1 & ATL083-10-06-72865-3 & ATL083-04-26-71638-2. (F.O.S. 4:1)
***Note**: These values only indicate Static Loads when the Caster locked.*

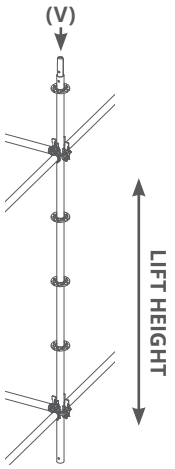


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STANDARDS

AXIAL LOAD			
LIFT HEIGHT		AXIAL LOAD (V)	
ft	m	lb	kN
3'-3"	1.00	7520	33.50
4'-11"	1.50	7395	32.90
6'-6"	2.00	4710	21.00

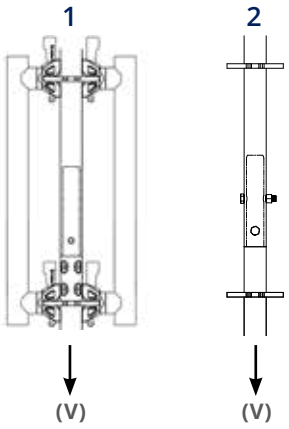
Test Reports: ATL083-03-01-78369-4, ATL083-04-20-98914-1, ATL083-04-20-98914-2. (F.O.S. 4:1)
***Note**: Diagonal Brace must span between nodes.*



SUSPENSION LOADS

TENSION LOAD			
DESCRIPTION		TENSION LOAD (V)	
		lb	kN
1	Leg Lock (Pair)	5236	23.30
2	Standard with Double Bolted Spigot	6250	27.80

Test Report: ATL083-05-01-99425-2 & ATL083-12-01-57407-1 (F.O.S. 4:1)
***Note**: Leg Locks MUST be used in pairs.*
Must use grade 8.8 or equivalent Bolt. Never use spigot pins to connect standards together for suspended applications. It is the user's responsibility the ensure the supporting structure is capable of supporting suspended scaffold loads.



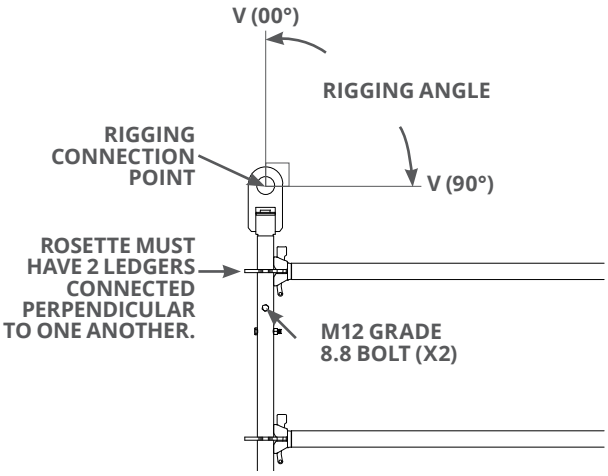
LIFTING LUG

LIFTING LUG		
ANGLE	LOAD	
Degrees	lb	kN
0	6250	27.8
45	2531	11.3
90	1260	5.6

***Note**: Top Rosette must be fully supported with 2 or more Ledgers.*
M12 Grade 8.8 Bolt (x2) must be used to secure Lug into Standard. It is recommended a spreader bar and long chains are used to ensure the correct angles are achieved as shown in the above table.
In the case where the lifting lug is loaded in two planes and at two different rigging angles simultaneously, the lesser Safe Working Load of the two angles in both planes shall be taken as the load capacity.

Interpolation is not recommended. When the rigging angle is not listed on the table, use the lesser SWL that the rigging angle falls in between.

Test Report: SHIN170200375CCM (F.O.S. 5:1)



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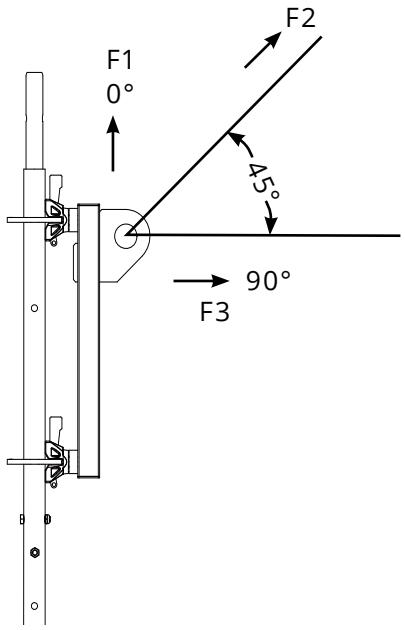
Lifting Lug

UNIVERSAL LIFTING LUG

UNIVERSAL LIFTING LUG	
ANGLE	LOAD
Degrees	(lbs)
0 (F1)	3543
45 (F2)	3743
90 (F3)	3378

Test Report: SHIN2308001219CM01_EN (F.O.S. 5:1)

Note: *Standard must be fully braced in two directions.*



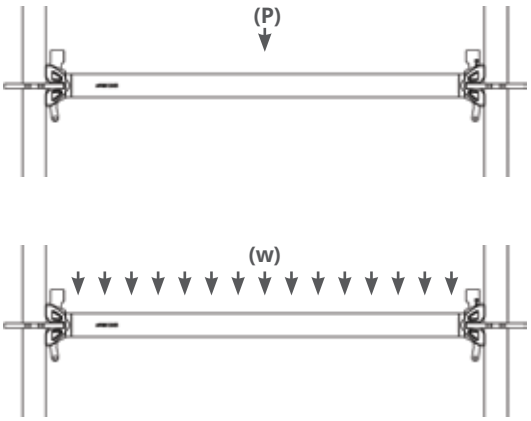
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LEDGERS O-TYPE

UNIFORM DISTRIBUTED LOAD AND POINT LOAD

LENGTH		UDL (w)		CENTER POINT LOAD (P)	
ft	m	lb/ft	kN/m	lb	kN
2'-2"	0.65	1749.50	26.00	3518	15.65
2'-10⅝"	0.88	1061.80	14.80	1730	7.70
3'-6"	1.065	594.30	8.70	1255	5.58
3'-10"	1.15	455.60	6.80	1180	5.25
5'	1.52	260.00	3.80	930	4.14
5'-2"	1.57	251.50	3.70	880	3.92
6'*	1.82*	180.70*	2.70*	776*	3.50*
7'	2.13	122.90	1.80	590	2.63
8'	2.43	94.40*	1.40*	485	2.16
10'	3.05	62.50	0.90	365	1.62

Test Report: ATL083-05-04-99575-1 & ATL083-12-23-45896-10; (F.O.S. 4:1)*
**Indicates calculated values derived from test values.*

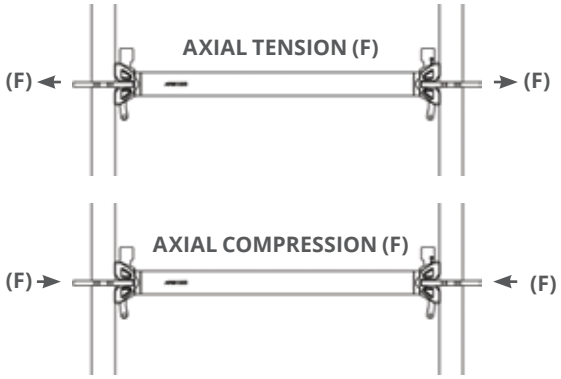


AXIAL COMPRESSION/TENSION LOAD

LENGTH		COMPRESSION (F)		TENSION (F)	
ft	m	lb	kN	lb	kN
2'-2"	0.65	4755	21.20	3375	15.00
2'-10⅝"	0.88	4755	21.20	3375	15.00
3'-6"	1.065	4755	21.20	3375	15.00
3'-10"	1.15	4665	20.80	3375	15.00
5'	1.52	4485	20.00	3375	15.00
5'-2"	1.57	4485	20.00	3375	15.00
6'	1.82	4040	18.00	3375	15.00
7'	2.13	3595	16.00	3375	15.00
8'	2.43	2445	10.90	3375	15.00
10'	3.05	1825	8.10	3375	15.00

Test Report: ATL083-01-06-46062-1 & ATL083-06-09-51804-1. (F.O.S. 4:1)
***Note**: 1. The safe axial compression loads do not make any allowance for service loads from working platforms that will apply bending and shear stresses to the Ledger tubes in addition to the stresses due to the axial compression. These combined stresses will reduce the safe axial compressive load capacity of the Ledgers. If in doubt, please check with Technical Services.*

2. The safe axial tensile loads do not make any allowance for service loads from working platforms that will apply bending and shear stresses to the Ledger tubes in addition to the stresses due to the axial compression. These combined stresses will reduce the safe axial tensile load capacity of the Ledgers. If in doubt, please check with Technical Services.

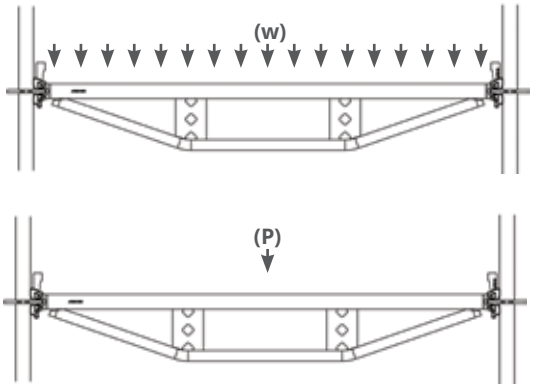


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TRUSS LEDGER O-TYPE

UNIFORM DISTRIBUTED LOAD AND POINT LOAD					
LENGTH		UDL (w)		CENTER POINT LOAD (P)	
ft	m	lb/ft	kN/m	lb	kN
5'-2"	1.57	1091	15.90	3475	15.46
6'	1.82	434	6.40	1910	8.50
7'	2.13	366	5.30	1350	6.01
8'	2.43	320	4.70	1855	8.25
10'	3.05	191	2.80	1065	4.74

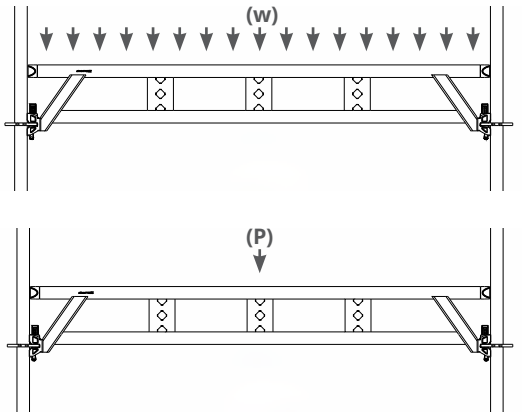
Test Report: ATL083-07-11-02718-1 & ATL083-02-23-47902-1. (F.O.S. 4:1)



RAISED TRUSS LEDGER

UNIFORM DISTRIBUTED LOAD AND POINT LOAD					
LENGTH		UDL (w)		CENTER POINT LOAD (P)	
ft	m	lb/ft	kN/m	lb	kN
5'-2"	1.57	995	14.4	2705	12.0
6'	1.82	861	12.6	2679	11.9
7'	2.13	750	11.0	2648	11.8
8'	2.43	599	8.8	2545	11.3
10'	3.05	385	5.6	2333	10.4

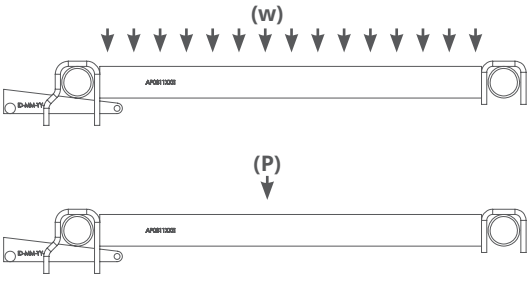
Test Report: 01-220-CHN-T-1901340 & 1901341 & 1901342 (F.O.S. 4:1)



MID TRANSOM

UNIFORM DISTRIBUTED LOAD AND POINT LOAD					
LENGTH		UDL (w)		CENTER POINT LOAD (P)	
ft	m	lb/ft	kN/m	lb	kN
5'-2"	1.57	111.50	1.63	494.80	2.20
6'	1.82	95.40	1.40	397.40	1.77
7'	2.13	67.50	0.99	345.40	1.54
8'	2.43	56.70	0.83	310.50	1.38
10'	3.05	41.20	0.60	241.50	1.07

Test Report: 1508708-001. (F.O.S. 4:1)

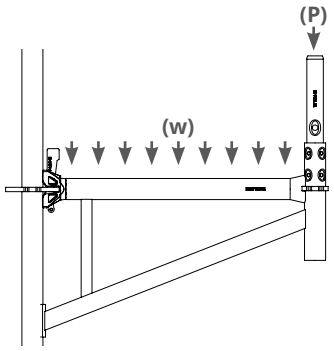


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SIDE/HOP-UP BRACKET O-TYPE

UNIFORM DISTRIBUTED LOAD AND POINT LOAD					
LENGTH		UDL (w)		POINT LOAD (P)	
ft	m	lb/ft	kN/m	lb	kN
1'	0.31	595.00	8.50	595	2.65
2'-2"	0.65	1467.80	21.80	1610	7.16
2'-10⅝"	0.88	816.40	11.40	1270	5.65

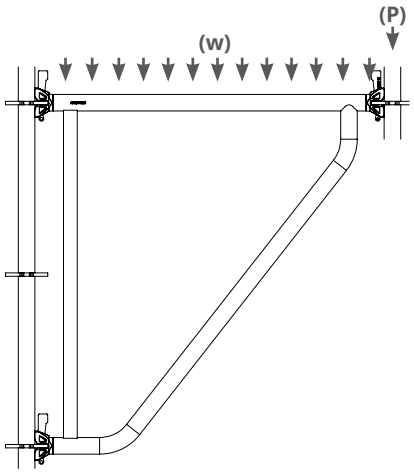
Test Report: ATL083-04-16-98536-1, ATL083-08-29-04888-1-4 & ATL083-12-23-45896-4. (F.O.S. 4:1)
***Note**: The safe working load of the Side Bracket may be limited by the load that is applied to the Standard to which the Side Bracket is affixed.*
Load conditions cannot be combined and must be applied independently.



SIDE/CONSOLE BRACKET O-TYPE

UNIFORM DISTRIBUTED LOAD AND POINT LOAD					
LENGTH		UDL (w)		POINT LOAD (P)	
ft	m	lb/ft	kN/m	lb	kN
3'-6"	1.065	750	11	1670	7.43
3'-10"	1.15	664.50	9.70	1870	8.32

Test Report:ATL083-08-29-04888-3 & ATL083-08-27-04888-4, ATL08301-01-26-46796-1. (F.O.S. 4:1)
***Note**: The safe working load of the Console Bracket may be limited by the load that is applied to the Standard to which the Console Bracket is affixed.*
Load conditions cannot be combined and must be applied independently.

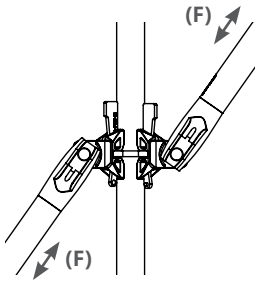


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BAY BRACE 6'-6" (2.0m) LIFT

TENSION AND COMPRESSION LOAD					
LENGTH		TENSION (F)		COMPRESSION (F)	
ft	m	lb	kN	lb	kN
2'-10⅝"	0.88	1890	8.40	1671	7.40
3'-6"	1.06	1890	8.40	1579	7.00
3'-10"	1.15	1890	8.40	1532	6.80
5'-2"	1.57	1890	8.40	1274	5.70
6'	1.82	1890	8.40	1125	5.00
7'	2.13	1890	8.40	968	4.30
8'	2.43	1890	8.40	830	3.70
10'	3.05	1890	8.40	618	2.70

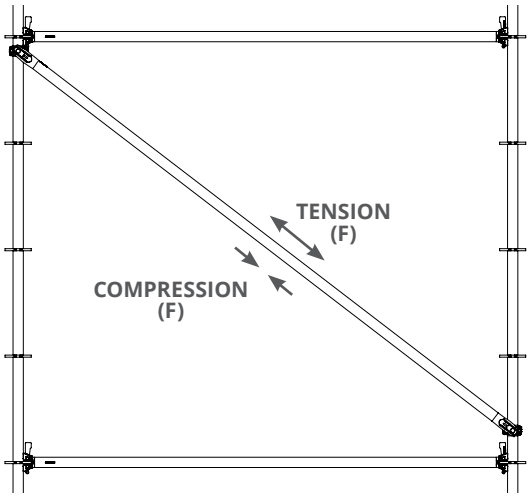
Test Report: ATL083-06-09-51804-1, PE Certified Capacity Tables AP-BBT-E & ATL083-12-23-45896-5. (F.O.S. 4:1)



SWIVEL CLAMP BRACE 6'-6" (2.0m) LIFT

TENSION AND COMPRESSION LOAD					
LENGTH		TENSION (F)		COMPRESSION (F)	
ft	m	lb	kN	lb	kN
5'	1.52	608	2.70	608	2.70
7'	2.13	608	2.70	608	2.70
8'	2.43	608	2.70	608	2.70
10'	3.05	608	2.70	588	2.60

Test Report: PE Certified Capacity Tables AP-ABT-E and ATL083-07-14-52956-1. (F.O.S. 4:1)



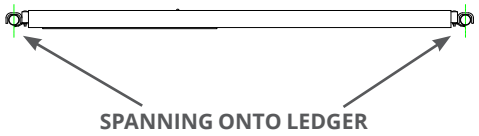
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Ladder Hatch Plank & Steel Plank

ALU/PLY LADDER HATCH PLANK

DUTY RATING			
ft	m	Duty Rating	lb/ft²
7'	2.13	Light Duty	25
8'	2.43	Light Duty	25
10'	3.05	Light Duty	25

Test Report: Ladder Hatch Plank 2.13m X 0.74m-Mk2-US-2018-01220CHN-T1801041 & Ladder Hatch Plank 3.05m x 0.74m-Mk2-US-2019. (F.O.S 4:1).



STEEL PLANK 9.5" (240mm) WIDE

UNIFORM DISTRIBUTED LOAD											
LENGTH (ft) (m)	2'-10⅝" 0.88	3' 0.914	3'-6" 1.065	3'-10" 1.15	4' 1.219	5' 1.52	5'-2" 1.57	6' 1.82	7' 2.13	8' 2.43	10' 3.05
UDL (lb/ft²)	620	395	305	250	228	182	176	152	133	98	69
UDL (kN/m²)	30	19	15	12	11	9	8	7	6	5	3
Light Duty	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Medium Duty	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Heavy Duty	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	

Test Report: ATL083-01-23-94071-1. (F.O.S. 4:1)

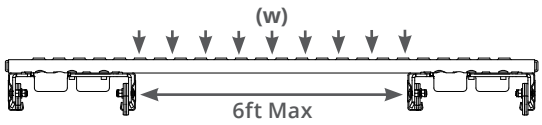


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GAP FILLER PLANK 320 & 190

GAP FILLER PLANK 320 & 190		
DESCRIPTION	L ≤ 6ft [1.829m]	DUTY RATING
UDL (psf)-320mm	34.6	Light Duty
UDL (psf)-190mm	63.8	Medium Duty

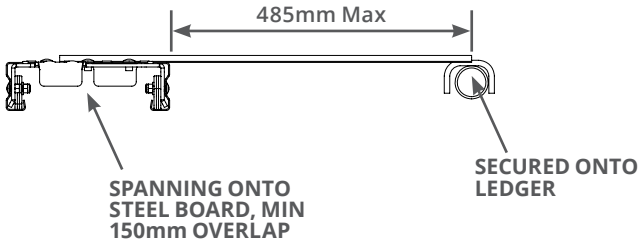
Gap Filler Planks require a minimum overlap of 6" [152mm] at each end.
Test Report: 01 220 CHN/T-2400115 (190mm) & 01 220 CHN/T-2400114 (320mm) (F.O.S. 4:1)



LAP PLATE – WITH HOOK

LAP PLATE – WITH HOOK		
DESCRIPTION	LAP PLATE – WITH HOOK	DUTY RATING
UDL (psf)	143	Heavy Duty

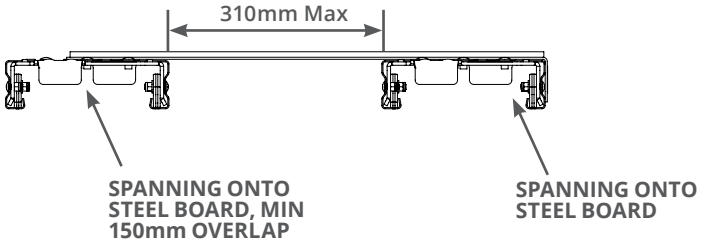
Test Report: 01 220 CHN/T-2305123 (F.O.S. 4:1)



LAP PLATE – WITHOUT HOOK

LAP PLATE – WITHOUT HOOK		
DESCRIPTION	LAP PLATE RA	DUTY RATING
UDL (psf)	1184	Heavy Duty

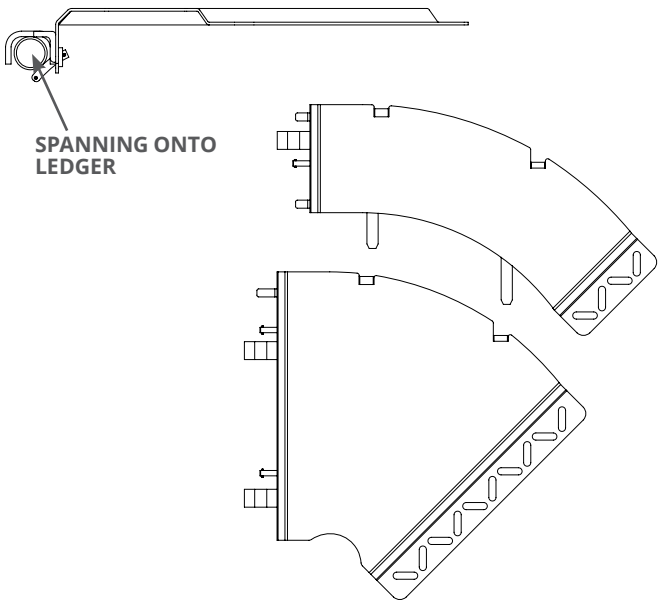
Test Report: 01 220 CHN/T-2305124 (F.O.S. 4:1)



RADIUS CORNER FILLER PLANK

RADIUS CORNER FILLER PLANK				
DESCRIPTION	INNER PLANK 596	OUTER PLANK 838	OUTER PLANK 1065	DUTY RATING
UDL (psf)	115	132	115	Heavy Duty

Test Report: Radiused corner filler plank testing report -20200109 (F.O.S. 4:1)

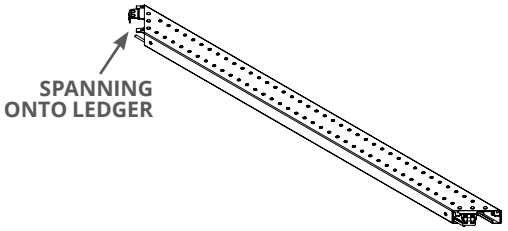


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CORNER PLANK

CORNER PLANK							
LENGTH (ft) (m)	1'-11" 0.60	2'-8 5/8" 0.84	3'-6" 1.065	4'-5" 1.34	5'-2" 1.57	6'-0" 1.82	7'-0" 2.13
Light Duty (25 psf)	✓	✓	✓	✓	✓	✓	✓

Reference Calculation: CA-10-02 45 Deg Corner

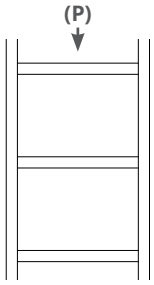


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STEEL VERTICAL LADDER

VERTICAL FORCE		
LENGTH		LOAD (P)
ft	m	
3'	0.91	*1) At least two loads of 250 lb (114 kg) each, concentrated between any two consecutive attachments.*
5'	1.52	*2) Each step or rung shall be capable of supporting a single concentrated load of at least 250 lb (144 kg) applied in the middle of the step or rung. (As per OSHA requirements)*

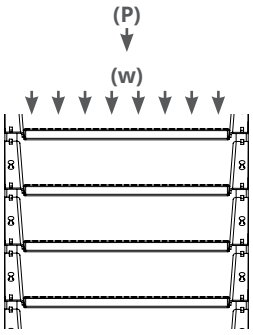
Test report No: ATL083-03-14-96867-1.



STEEL STAIR STRINGER AND TREAD

UNIFORM DISTRIBUTED LOAD AND POINT LOAD ON TREAD					
BAY HEIGHT		UDL (w)		CENTER POINT LOAD (P)	
ft	m	lb/ft²	kN/m²	lb	kN
6'-6"	2.00	50	2.40	337	1.50

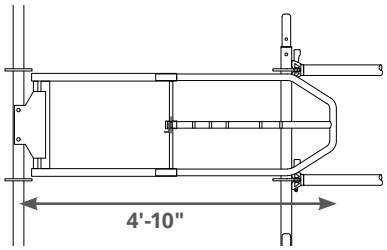
Test Report: ATL083-10-23-07476-1 and ATL083-03-18-60358-1; (F.O.S. 4:1)
***Note**: Load conditions cannot be combined and must be applied independently.*



SAFETY GATE

VERTICAL/HORIZONTAL LOADS		
LENGTH		LOAD CONDITION
ft	m	
3'-1" – 4'-10"	0.94 – 1.48	*The Adjustable Safety Gate is capable of withstanding, without failure, a force applied in any vertical or horizontal direction at any point along its top edge of 200lb (as per OSHA requirements).*

Test Report: ATL083-03-10-60131-1.

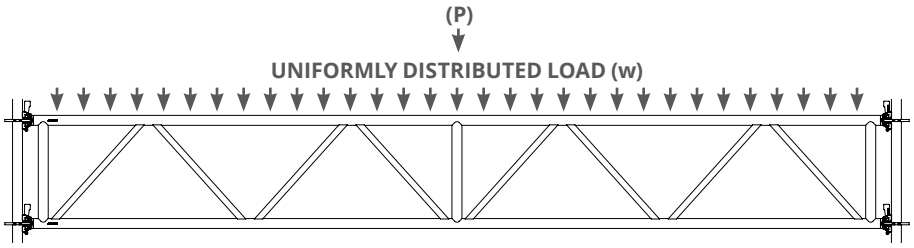


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LATTICE GIRDER, 19.7" (0.5m)

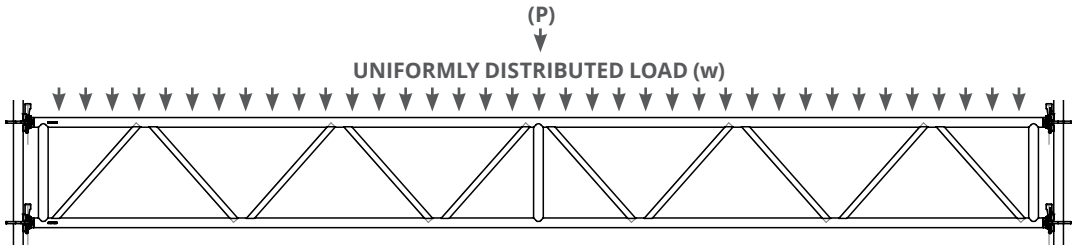
UNIFORM DISTRIBUTED LOAD AND POINT LOAD (14’)							
LENGTH		TOP CHORD BRACING CONDITIONS (C/C)		UDL (w)		CENTER POINT LOAD (P)	
ft	m	ft	m	lbs/ft	kN/m	lb	kN
14’	4.26	1'	0.308	168.75	2.47	2120	9.43
		3'-4”	1.02	168.75	2.47	2120	9.43
		7'	2.13	141.25	2.07	1000	4.45

Test Report: 2015-095 & 2015-137. (F.O.S. 4:1)
***Note**: Load conditions cannot be combined and must be applied independently. Ilustration show Mk2 Lattice Girder without spigot.*



UNIFORM DISTRIBUTED LOAD AND POINT LOAD (17’)							
LENGTH		TOP CHORD BRACING CONDITIONS (C/C)		UDL (w)		CENTER POINT LOAD (P)	
ft	m	ft	m	lbs/ft	kN/m	lb	kN
17’	5.18	1'	0.308	136.25	1.99	1950	8.68
		3'-4”	1.02	136.25	1.99	1810	8.05
		7'	2.13	94.50	1.38	823	3.66

Test Report: 2015-095 & 2015-137. (F.O.S. 4:1)
***Note**: Load conditions cannot be combined and must be applied independently. Ilustration show Mk2 Lattice Girder without spigot.*

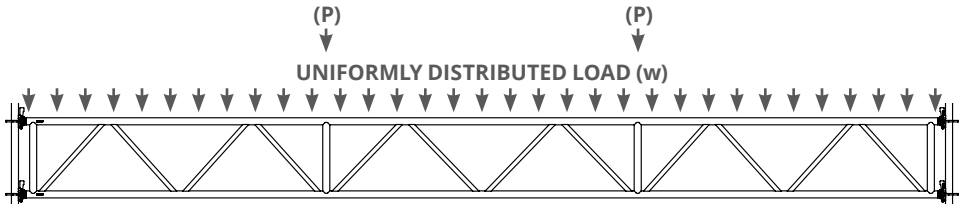


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LATTICE GIRDER, 19.7" (0.5m) CONTINUED

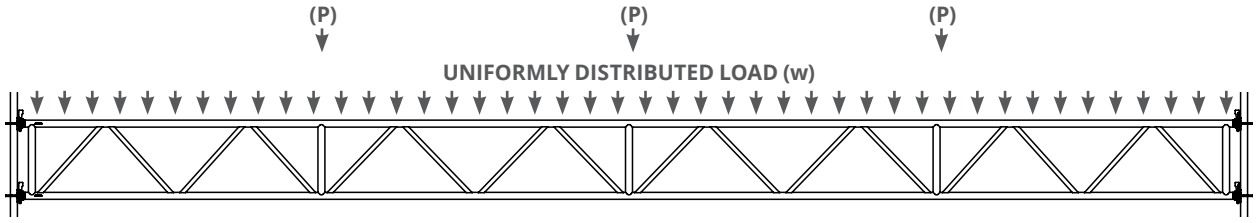
UNIFORM DISTRIBUTED LOAD AND POINT LOAD (21’)							
LENGTH		TOP CHORD BRACING CONDITIONS (C/C)		UDL (w)		POINT LOAD (P) (7’ C/C)	
ft	m	ft	m	lbs/ft	kN/m	lb	kN
21’	6.40	1'	0.308	107.50	1.57	1055	4.69
		3'-4”	1.02	107.50	1.57	1055	4.69
		7'	2.13	64.50	0.94	503	2.24

Test Report: 2015-095 & 2015-137. (F.O.S. 4:1)
Note: *Load conditions cannot be combined and must be applied independently. Illustration show Mk2 Lattice Girder without spigot.*



UNIFORM DISTRIBUTED LOAD AND POINT LOAD (28’)							
LENGTH		TOP CHORD BRACING CONDITIONS (C/C)		UDL (w)		POINT LOAD (P) (7’ C/C)	
ft	m	ft	m	lbs/ft	kN/m	lb	kN
28’	8.52	1'	0.308	79	1.16	697.5	3.10
		3'-4”	1.02	79	1.16	658	2.83
		7'	2.13	37.5	0.55	263	1.17

Test Report: 2015-095 & 2015-137. (F.O.S. 4:1)
Note: *Load conditions cannot be combined and must be applied independently. Illustration show Mk2 Lattice Girder without spigot.*



Note: *The most common failure of trusses is due to inadequate lateral bracing. When such lateral bracing is not installed properly, the trusses undergo twisting and rotation that can result in lateral torsional buckling. Lateral bracing shall be used to stabilize the top chord of the truss against lateral displacement and truss rotation to prevent such failure.*

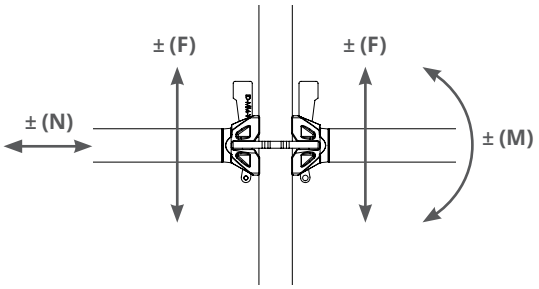
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Rosette Clamp, Davit Arm, & Wall Tie Bracket

ROSETTE CLAMP

MOMENT, NORMAL AND VERTICAL LOAD					
CONNECTION MOMENT (M)		NORMAL LOAD (N)		VERTICAL LOAD (F)	
lb.ft	kNm	lb	kN	lb	kN
±205	0.28	±2435	10.84	±1460	6.50

Test Report: ATL083-11-08-26890-1 (F.O.S 4:1)
***Note:** Bolt tightened to 38ft•lb*



L TYPE DAVIT ARM

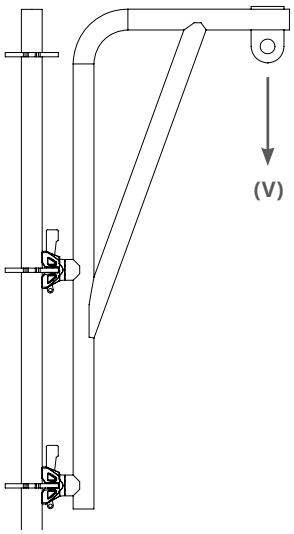
LOAD FOR LIFTING		
DESCRIPTION	VERTICAL LOAD (V)	
	lb	kN
L Type Davit Arm	1075	4.78

Test Report: ATL083-12-07-62651-1. (F.O.S. 4:1)

MAX SHOCK LOAD (V)		
DESCRIPTION	SHOCK LOAD (V)	
	lb	kN
L Type Davit Arm	5000	22.20

Test Report: ATL083-06-18-52151-2.
***Note:** Please refer to Safety Harness Connection Guide for more information regarding use of the Davit Arm as a connection point for fall protection.*

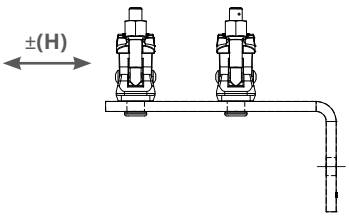
The Standard the Davit Arm is connected to, must be supported in two directions at the top connection of the Davit Arm. Ringlock components or Tube and Clamp can be used for bracing. All joints between Standards should be pinned together.



WALL TIE BRACKET

WALL TIE BRACKET		
SLIP LOAD		
CONNECTION	H (lb)	H (kN)
2 Clamp	1294	5.75

Test Report: SHCCM160602325 (F.O.S. 4:1)



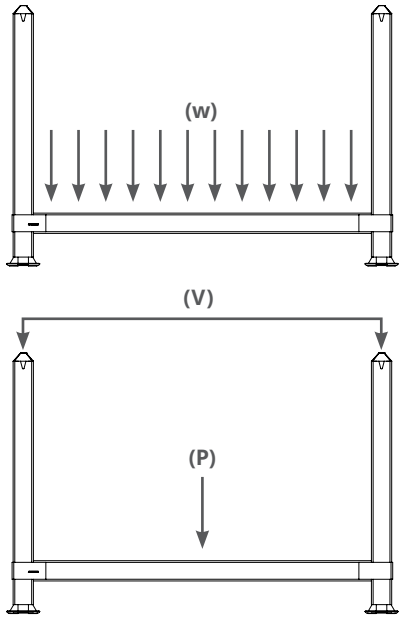
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RINGLOCK RACK

UNIFORM DISTRIBUTED LOAD, CENTER POINT LOAD AND AXIAL LOAD					
TOTAL UNIFORM LOAD (W)		TOTAL POINT LOAD (P)		TOTAL AXIAL LOAD (V)	
lb	kN	lb	kN	lb	kN
9642	42.90	5765	25.70	12902	57.41

Test Report: ATL083-03-05-96410-1 (F.O.S 4:1)
***Note**: Axial Load assumes all 4 legs are loaded simultaneously and uniformly.*

When racks are loaded, ground condition must be assessed to withstand the load. Use soleboards (mudsills) as required.

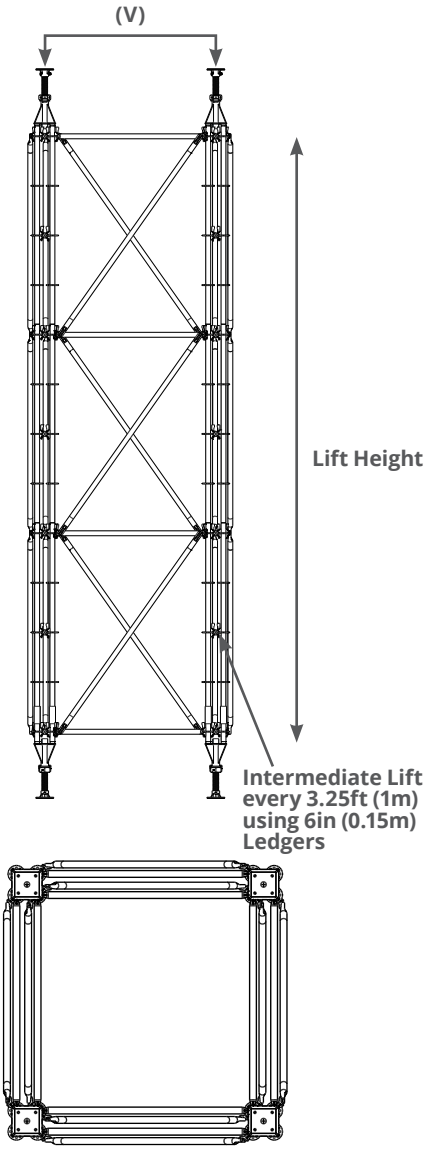


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RINGLOCK HD SHORING CAPACITIES

RINGLOCK HD SHORING CAPACITIES				
PLAN SIZE	WIND LOAD	TOTAL TOWER LOAD (V) KIPS 6'-6" (2m) LIFTS WITH 6" (0.15m) LEDGER AT 3'-3" (1m) @ 2.5 SAFETY FACTOR		
		1 Lift (6.5ft)	2 Lifts (13ft)	3 Lifts (19.5ft)
5'-2" x 5'-2" (1.57m x1.57m)	No Wind	107.15	95.92	90.85
	4psf Wind	106.61	95.12	89.78
	16psf Wind	104.21	92.99	87.37
5'-2" x 7'-0" (1.57m x2.13m)	No Wind	105.28	95.12	88.98
	4psf Wind	104.21	94.59	87.91
	16psf Wind	102.07	92.18	84.97
5'-2" x 10'-0" (1.57m x3.05m)	No Wind	105.28	95.12	88.98
	4psf Wind	104.21	94.59	87.91
	16psf Wind	102.07	92.18	84.97
7'-0" x 7'-0" (2.13m x2.13m)	No Wind	107.41	97.53	91.38
	4psf Wind	106.88	96.46	90.31
	16psf Wind	104.21	94.05	87.64
7'-0" x 10'-0" (2.13m x3.05m)	No Wind	107.41	97.53	91.12
	4psf Wind	106.35	96.46	90.05
	16psf Wind	104.21	93.79	86.57
10'-0" x 10'-0" (3.05m x3.05m)	No Wind	109.02	98.33	91.38
	4psf Wind	108.22	98.06	90.31
	16psf Wind	105.54	97.26	87.64

***Note**: Reference notes on page 26.*
Reference Calculation: 217047 (F.O.S 2.5:1)



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Ringlock Shoring

SHORING COLUMN CONSITING OF 4 STANDARDS CONNECTED BY 6" (150mm) X 6" (150mm) AT 3'-3" (1m)	
COLUMN LIFT HEIGHT	SWL POINT LOAD AT 2.5 SF
	kip
6.5ft (2m)	26.39
13ft (4m)	6.68
19.5ft (6m)	2.81

Reference Calculation: 217047 (F.O.S 2.5:1)

Note:

1. Shoring towers are to be erected in accordance with the drawings and all components are assumed to be in good condition. All members including bracing and horizontals are critical to the stability of the shoring tower.

2. The supported structure is assumed to be laterally restrained/stable. Safe working loads are based on a stable structure where the shoring tower is laterally restrained in both horizontal directions at the top.

3. Applied vertical loads are assumed to be applied equally at the top via Heavy Duty Screwjack with 4 spigots and distributed equally to each standard at each main upright (each main upright has 4 standards). The standards at each main upright are connected together via 6" Ledgers at every 3.25ft increment.

4. Shoring towers are assumed to be supported on a stable and level surface. The reaction load at each main upright is the total load divided by four.

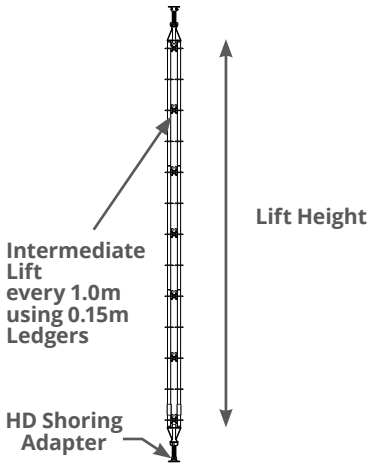
5. A horizontal load of 2% of the vertical load is applied to the shoring tower to account for imperfections and any eccentric loadings.

6. Wind loads are considered at 4psf for conditions with minimal exposure to wind and 16psf for moderate to extreme exposure to wind. If the design wind load is known to be more than 16psf, this table does not apply and needs to be evaluated by an engineer.

7. The loads stated are safe working loads at a safety factor of 2.5. For conversion of the ultimate limit state multiply the safe working loads by 2.5.

8. The shoring towers are assumed to be open and exposing only the scaffolding components to wind loads.No cladding, netting, or hoarding is to be added to the shoring towers.

9. Structural analysis is in accordance with the requirements of AISC 15th (360-16) ASD.



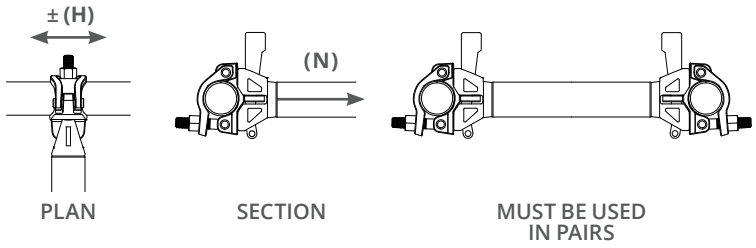
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Rosette and Adapter Clamps

HORIZONTAL ROSETTE CLAMP

HORIZONTAL ROSETTE CLAMP	
AXIAL (N)	SLIP (H)
lb	lb
3422	811

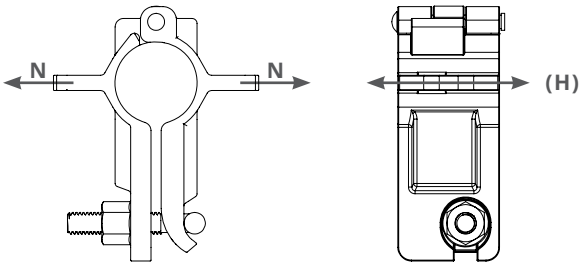
Test Report: 01 220 CHN/T-2305133 (F.O.S. 4:1)



DOUBLE ROSETTE CLAMP

DOUBLE HORIZONTAL ROSETTE CLAMP	
AXIAL (N)	SLIP (H)
lbs	lbs
2318	1485

Test Report: 01 220 CHN/T-2302783 (F.O.S. 4:1)

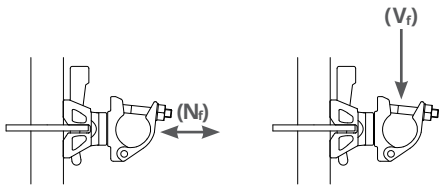


SWIVEL AND RA ADAPTOR CLAMP

SWIVEL AND RA ADAPTOR CLAMP		
DESCRIPTION	AXIAL (Nf)	SHEAR (Vf)
	lbs	lbs
SWIVEL	2388	1044
RIGHT ANGLE	3514	1603

***Note**: when load is applied at the tube end, it is both subjugated to bending and shear.*

Test Report: 01 220 CHN/T-2305131, 01 220 CHN/T-2305132 & 01 220 CHN/T-2305128 (F.O.S. 4:1)



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TUBE & CLAMP COMPONENT SAFE WORKING LOADS

The following values and the values throughout are Safe Working Loads, these are derived by testing components to failure and adding a **F.O.S of 4:1**, i.e. Ultimate Loads divided by 4. For more information please consult your local AT-PAC representative.

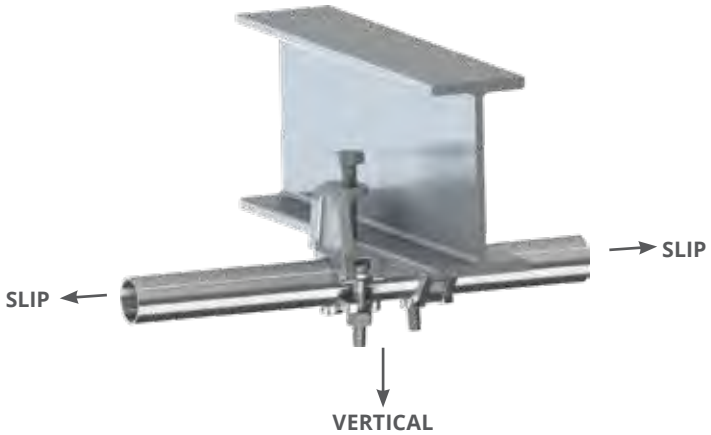
CLAMPS

I-BOLT AND WEDGE CLAMPS			
COMPONENT	SLIP lb	VERTICAL lb	TEST REPORT NO.
D/F Right Angle Clamp	1415	-	ATL083-03-26-49324-1
D/F Right Angle Clamp with (1) Check Clamp***	2100	-	ATL083-03-26-49324-1
D/F Swivel Clamp♦	1040	-	ATL083-10-11-25216-2
D/F Beam Clamp Rigid*	705	4000	ATL083-12-02-57437-1 & ATL083-06-21-68812-1
D/F Beam Clamp Swivel*	705	2940	ATL083-12-02-57437-1 & ATL083-01-04-76097-1
Right Angle Wedge Clamp	1530	-	ATL083-03-25-14695-2
Right Angle Wedge Clamp**	1170	-	ATL083-03-25-14695-1
Swivel Wedge Clamp	1355	-	ATL083-06-24-19860-1

Notes: *All Clamps are to be tightened to 40 ft. lb torque. Clamps tested in accordance to ANSI/SSFI SC100-5/05 and EN-74 test methods*
Vertical force is only applicable to the Beam Clamps.
**Must be used in pairs.*
***Attached to Aluminum Tube. F.O.S. 4:1*
****The capacity for additional Check Clamps is 50% of the original clamp.*
♦The limiting factor of Swivel Clamp is the center rivet (pin connection); therefore, you cannot increase slip capacity with a Check Clamp.



I-BOLT SLIP LOAD

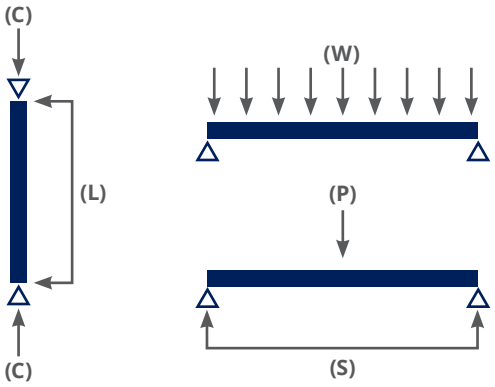


BEAM CLAMP SLIP LOAD

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STEEL TUBE (WITH FITTING)

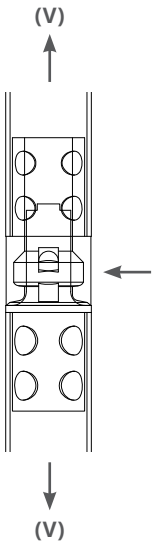
STEEL TUBE 1.9" X 0.098" (48.3mm X 2.5mm)			
EFFECTIVE LENGTH (S) & (L)	AXIAL COMPRESSION (C)	POINT LOAD (P)	UDL (W)
ft	lb	lb	lb/ft
1'	4668	910	1820
2'	4428	455	455
3'	4055	303	202
4'	3585	227	114
5'	3059	182	73
6'	2521	152	51
7'	2005	130	37
8'	1548	114	28
9'	1223	101	23
10'	990	91	18



Notes: *All loads stated are safe working (F.O.S. 4:1).*
Load Conditions must not be combined and act independently.
*Safe Working Bending Moment ϕM_n = **227.46** ft•lbs*
*Safe Working Shear Force ϕV_n = **1425.27**lb*
Deflection is L/180 or better
Table based on simply supported beam with connections restraining torsional rotation.
Material Specification ASTM A283 Grade C or Equivalent

TENSION LOAD	
DESCRIPTION	TENSION LOAD (V)
	lb
Steel Tube With Fitting	4751

It is the user's responsibility to ensure the support structure is capable of supporting suspended scaffold loads (F.O.S. 4:1)



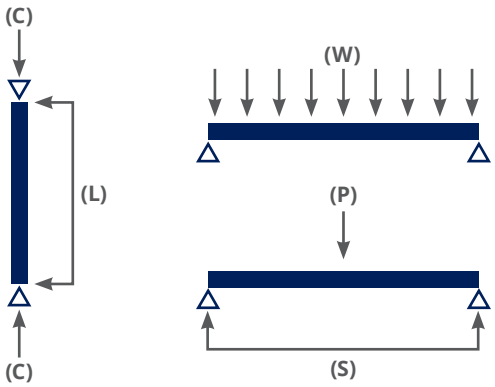
JOINT BETWEEN STEEL TUBE WITH FITTING

***Note**: When used in tension, joint must be spliced together using Tube & Clamp.*

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STEEL TUBE (OPEN ENDED)

STEEL 1.9" OD X 0.125" (48.3mm x 3.2mm)			
EFFECTIVE LENGTH (S) & (L)	AXIAL COMPRESSION (C)	POINT LOAD (P)	UDL (W)
ft	lb	lb	lb/ft
1'	8802	1707	3414
2'	8108	854	854
3'	7071	569	379
4'	5838	427	213
5'	4563	341	137
6'	3369	285	95
7'	2476	244	70
8'	1895	213	47*
9'	1498	185*	33*
10'	1213	149*	24*

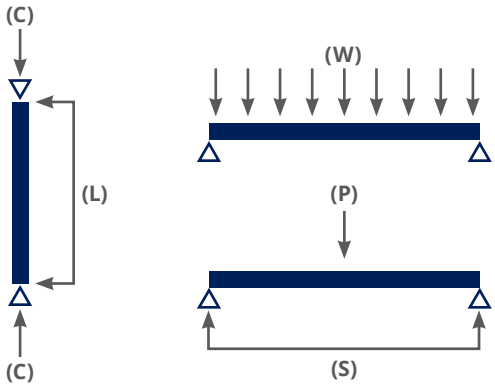


Notes: *All loads stated are safe working (F.O.S.4:1)*
Load Conditions must not be combined and act independently.
*Safe Working Bending Moment ϕM_n = **426.77**ft•lbs*
*Safe Working Shear Force ϕV_n = **2713.81**lb*
*Deflection is L/180 or better. *Denotes deflection governed.*

Table based on simply supported beam with connections restraining torsional rotation.
Material Specification S355J0H, EN 10219-1 or equivalent.

ALUMINUM TUBE (OPEN-ENDED)

ALUMINUM 1.9" X 0.145" (48.3mm X 3.7mm)			
EFFECTIVE LENGTH (S) & (L)	AXIAL COMPRESSION (C)	POINT LOAD (P)	UDL (W)
ft	lb	lb	lb/ft
1'	6210	1060	2125
2'	5030	530	530
3'	3960	355	235
4'	2560	265	135
5'	1640	215	75
6'	1140	160	45
7'	840	120	25
8'	640	90	18
9'	510	70	12
10'	410	60	9



Notes: *All loads stated are safe working (F.O.S. 4:1).*
Load Conditions must not be combined and act independently.
*Safe Working Moment ϕM_n = **225** ft•lbs*
*Safe Working Shear Force ϕV_n = **1710**lb*
Deflection is L/180 or Better.

Table based on simply supported beam with connections restraining torsional rotation.
Material Specification Alloy 6061 T6

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ALUMINUM BEAM 6.5” (165mm)

ALUMINUM 6.5” BEAM (165mm)								
EFFECTIVE LENGTH SPAN	ALLOWABLE DEFLECTION		1 SPAN		2 SPAN		3 SPAN	
ft	in		lb/ft		lb/ft		lb/ft	
4’	0.13		2947 M		2140 R		2432 R	
4’-6”	0.15		2328 M		1902 R		2162 R	
5”	0.17		1886 M		1712 R		1945 R	
5’-6”	0.18		1514 D		1556 R		1769 R	
6’	0.20		1166 D		1310 M		1621 R	
6’-6”	0.22		917 D		1116 M		1395 R	
7’	0.23		734 D		962 M		1203 M	
7’-6”	0.25		597 D		838 M		1048 M	
8’	0.27	0.25	492 D	461*	737 M		921 M	872*
8’-6”	0.28	0.25	410 D	362*	653 M		776 D	685*
9’	0.30	0.25	346 D	288*	582 M		654 D	545*
9’-6”	0.32	0.25	294 D	288*	522 M		556 D	439*
10’	0.33	0.25	252 D	189*	471 M	455*	476 D	357*
10’-6”	0.35	0.25	218 D	155*	428 M	374*	412 D	294*
11’	0.37	0.25	189 D	129*	390 M	311*	358 D	244*
11’-6”	0.38	0.25	166 D	108*	357 M	260*	313 D	204*
12’	0.40	0.25	146 D	91*	327 M	219*	276 D	172*
12’-6”	0.42	0.25	129 D	77*	302 M	186*	244 D	146*

Notes: *All loads stated are safe working (F.O.S. 2.2:1).*
***R** Denotes reaction governed. With beam fully supported on 8” (200mm) head.*
***V** Denotes shear governed.*
***M** Denotes bending moment governed.*
***D** Denotes deflection governed (deflection <L/360).*
***** Denotes deflection governs with deflection limited to 0.25" (6mm)*

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